Degassing and O2 Measurement with sensor cap with air stable substances

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Category: Protocol
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Goal

Degassing and O2 measurement of air stable materials with the sensor cap

Prerequisites and preparation

- 1 Schlenk
- 1 5 mL-Schlenk
- PTFE canula
- 2 septum
- Equipment - NS14 Schlenk tube for sensor cap
- [Equipment] Measurement Laptop
- Equipment - Firesting Fiber-Opitc Oxygen Meter 2 Channel

Steps

Step number	Step description
1	The 5 mL-Schlenk flask is equipped with a stopcock and the sensor cap and a glas or micro stir bar and the solution/suspension
2	If not available a 10 mL Schlenk flask is equipped with a stopcock, a septum containing a small PTFE-tube and a second septum on the PTFE tube
3	The second septum is put on the stopcock end of the 5 mL Schlenk flask.
4	The sensor cap is equipped with the glas fiber cables. The cables are connceted to the Equipment - Firesting Fiber-Opitc Oxygen Meter 2 Channel, which is connected to the Laptop
5	The Program PyroScience Workbench and the Logging is started.
6	The solution is degassed by bubbling Argon through the solution. Therefor the PTFE-tube is placed in the solution (the tube should be bent at the end, so that the tube can reach the solution/suspension. The solution/suspension is thereby stirred stongly (approx. 1000 rpm)
7	The solution is degassed till the oxygen concentration is constant (approx. 20 min), max. O2: 0.025 %. The value is noted and the Logging is stopped
8	The PTFE tube is removed from the solution. and the 5 mL-Schlenk is closed against the enviroment.
9	The measurment equippment (glas fibers, PyroScience and Laptop) is removed, when the Schlenk has to be moved. Otherwise continue with step 12
10	The 5 mL-Schlenk is moved to the Photochem. setup which has been setup according to Protocol - Irradiation setup

11	The measurment equppment (glas fibers, PyroScience and Laptop) is connected again and the Logging is started. The O2 conectration is noted.
12	With active Log the reaction is allowed to equlibrate for approx. 20 min.
13	The irradiation is started, without chainging anything else. Time is noted in seconds from start of the Log
14	After enough data is collected (typically more than 20 min) the irradiation is stopped without changing anything else and the time is noted in seconds from start of the Log
15	The Logging is continued for another approx. 15 to 20 min.
16	The measurment equippment is removed and the reaction mixture can be used for further experiments, after removing the sensor cap (in Argon counterflow if neccessary)

Linked resources

Equipment - GL14 Schlenk tube with sensor spots

Equipment - NS14 Schlenk tube for sensor cap

Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 1)

Protocol - Irradiation setup

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